The Domain of Hyperreal Differentiation Is Always Proper

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Abstract

Alexander and Dawson define differentiation on those elements of an ultrapower of the reals which have a representative differentiable on an ultrafilter-large set. For an additive idempotent ultrafilter concentrated immediately to the right of zero, the derivative is independent of the chosen differentiable representative. They ask whether its domain can be the whole ultrapower, and whether the answer depends on the ultrafilter. We prove in ZFC that the domain is proper for every such ultrafilter. More strongly, for each ultrafilter there is a bounded strictly positive function such that every equivalent representative is discontinuous on an ultrafilter-large set. A dichotomy for the rational coordinates of a Hamel basis supplies the construction. The resulting excluded element is a positive infinitesimal, so the differentiation domain is not an order-convex subfield. The witness may depend on the ultrafilter; no measurability or common-witness assertion is made.

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1 A derivative of a hyperreal number

An ultrapower represents a number by a real-valued function. This makes it tempting to differentiate the function and regard the resulting class as the derivative of the number. The difficulty is that many functions represent the same number. Agreement on a large set need not imply agreement of derivatives, and an equivalence class may contain both well-behaved and badly behaved functions.

Alexander and Dawson [1] identify an algebraic condition on the ultrafilter which resolves the first difficulty. If the ultrafilter is additively idempotent and concentrates near zero from the

right, any two equivalent representatives whose derivatives exist on large sets have equivalent derivatives. Their Corollary 10 therefore defines a derivative on a distinguished domain $\mathcal{D}_p$. Immediately afterwards they ask whether every hyperreal has a representative of the required kind, and whether this depends on p. Admissible ultrafilters exist in ZFC, as recorded in their Lemma 7; the universal statement below is therefore nonvacuous.

There are two different questions here. It is easy to take a good representative and alter it away from a large set so that it becomes badly behaved. That does not exclude its equivalence class from $\mathcal{D}_p$: the original good representative still exists. To prove properness one must construct a class for which *every* representative fails. The proof below is designed around that quantifier.

Theorem 1.1 (Universal properness)**.** *Let p be an additive idempotent ultrafilter on all subsets of $\mathbb{R}$ such that $(0,\epsilon) \in p$ for every $\epsilon > 0$. There is a function $f : \mathbb{R} \to (0,1]$ with the following properties.*

1. *For every $\eta > 0$, $\{x : f(x) < \eta\} \in p$.*
2. *For every $g : \mathbb{R} \to \mathbb{R}$ with $\{x : g(x) = f(x)\} \in p$, there is a set $S_g \in p$ at every point of which g is discontinuous.*

Consequently $[f]_p$ is a positive infinitesimal outside $\mathcal{D}_p$, and $\mathcal{D}_p \neq \mathbb{R}^{\mathbb{R}}/p$.

The first assertion places the excluded element very close to zero in the order of the ultrapower. The second is the essential obstruction: it survives all changes of representative. The proof uses finite intersections of large sets only. It never asks an ultrafilter to be closed under countably many intersections.

2 Ultrafilters, addition and the differentiation domain

An ultrafilter p on a set X is a family of subsets closed under supersets and finite intersections, excluding the empty set, and containing exactly one of A and $X \setminus A$ for each $A \subseteq X$. Call its members *p-large*. It is principal at x if its members are exactly the sets containing x; otherwise it contains no finite set.

For subsets of an additive group, put $A - x = \{y : x + y \in A\}$. Addition of ultrafilters on the underlying discrete group is defined by

$$A \in p + q \quad \iff \quad \{x : A - x \in q\} \in p.$$

Thus idempotence $p + p = p$ says that whenever A is large, a large set of translations has a large intersection pattern with A:

$$A \in p \quad \implies \quad A^* := A \cap \{x : A - x \in p\} \in p. \tag{2.1}$$

Near-zero concentration implies that p is nonprincipal: no single real belongs to every interval $(0,\epsilon)$.

Two functions $f, g : \mathbb{R} \to \mathbb{R}$ represent the same element of $\mathbb{R}^{\mathbb{R}}/p$ exactly when $\{x : f(x) = g(x)\} \in p$. Addition, multiplication and order are defined pointwise modulo p. The usual ultraproduct argument gives an ordered field extending $\mathbb{R}$ through constant functions.

Let D_f be the set of points where f has an ordinary finite two-sided derivative. The class $[f']_p$ exists when $D_f \in p$: extend the partial derivative arbitrarily outside D_f. The extension does not affect its class. The source domain is

$$\mathcal{D}_p = \{[f]_p : D_f \in p\}. \tag{2.2}$$

Membership means that *some* representative has this property. Merely having differentiability points arbitrarily close to zero is weaker than $D_f \in p$ and does not replace this definition.

For context we give the argument behind the well-definedness result of [1, Theorem 9].

Proposition 2.1. *If $[f]_p = [g]_p$ and $D_f, D_g \in p$, then $[f']_p = [g']_p$. Moreover, $\mathcal{D}_p$ is a subfield of the ultrapower and $\partial[f]_p = [f']_p$ is a derivation from it to the ultrapower.*

Proof. Let $A = \{f = g\} \cap D_f \cap D_g$. By (2.1), $A^* \in p$. Fix $x \in A^*$. For every $\delta > 0$, the set $(A - x) \cap (0, \delta)$ is large and hence nonempty. Along the arbitrarily small positive increments in this set, the difference quotients for f and g at x are equal, since both x and $x + h$ lie in A. Both ordinary derivatives exist, so their values agree. They therefore agree on A^*, proving the first assertion.

Constants are in the domain. Intersecting differentiability sets and applying the ordinary sum and product rules proves closure under sums, negatives and products and proves the derivation laws. If $[f]_p \neq 0$, then $\{f \neq 0\} \in p$. Define $h(x) = 1/f(x)$ where $f(x) \neq 0$ and $h(x) = 0$ otherwise. At every point of $D_f \cap \{f \neq 0\}$, continuity of f makes this the ordinary reciprocal in a neighborhood. Hence h is differentiable there, $[h]_p = [f]_p^{-1}$, and the domain is a field. □

The proposition explains why the properness question is meaningful. It supplies no reason that every equivalence class has a suitable representative. The next lemma isolates what will prevent one.

3 An obstruction which survives a change of representative

Lemma 3.1. *Suppose $f : \mathbb{R} \to (0, 1]$ has this property: whenever $T \in p$, $x \in T^*$ and $\delta > 0$, there exists*

$$y \in (T - x) \cap (0, \delta) \quad \text{such that} \quad f(x + y) < f(x)/2.$$

Then every g equivalent to f is discontinuous at every point of a p-large set. In particular $[f]_p \notin \mathcal{D}_p$.

Proof. Take $T = \{g = f\} \in p$. At each fixed $x \in T^*$, $g(x) = f(x) > 0$. The hypothesis gives arbitrarily small positive y with $x + y \in T$ and $g(x + y) = f(x + y) < g(x)/2$. The fixed positive tolerance $g(x)/2$ witnesses discontinuity at x. Since $T^* \in p$, the continuity set of g is not large: otherwise its intersection with T^* would be empty and large. Differentiability implies continuity, so its differentiability set cannot be large either. The argument applies to every representative. □

The use of idempotence is visible here. Ordinary equality modulo p only gives the set T. Idempotence provides points $x \in T$ at which the increments leading back into T remain large. Near-zero concentration then turns those increments into a continuity test.

4 Rational coordinates and the two cases

Choose a Hamel basis B for $\mathbb{R}$ as a vector space over $\mathbb{Q}$. For every $x \in \mathbb{R}$ there is a unique finite expression

$$x = \sum_{b \in B} h_b(x) b, \qquad h_b(x) \in \mathbb{Q}.$$

Each coordinate $h_b : \mathbb{R} \to \mathbb{Q}$ is additive. Write

$$\operatorname{supp}(x) = \{b : h_b(x) \neq 0\}, \qquad \ell(x) = |\operatorname{supp}(x)|.$$

The support is finite; $\ell(x) = 0$ precisely when $x = 0$. Disjoint supports give $\ell(x+y) = \ell(x) + \ell(y)$.

For a function $h : X \to Y$, its pushforward ultrafilter is $h_* p = \{A \subseteq Y : h^{-1}(A) \in p\}$.

Lemma 4.1. *If $h : \mathbb{R} \to \mathbb{Q}$ is additive, $h_* p$ is an additive idempotent ultrafilter on $\mathbb{Q}$. If it is principal, it is principal at zero.*

Proof. Put $q = h_* p$. For $A \subseteq \mathbb{Q}$, additivity gives

$$\begin{aligned} A \in q + q &\iff \{x : h^{-1}(A - h(x)) \in p\} \in p \\ &\iff \{x : h^{-1}(A) - x \in p\} \in p \\ &\iff h^{-1}(A) \in p + p = p \iff A \in q. \end{aligned}$$

The sum of principal ultrafilters at r and r is principal at $2r$. Idempotence therefore implies $r = 2r$, hence $r = 0$. □

It follows that precisely one of the following alternatives holds: some $h_{b*} p$ is nonprincipal; or every $h_{b*} p$ is principal at zero. The second alternative says $\ker h_b \in p$ for every $b \in B$. It does not say the intersection of all these kernels is large. That intersection is just $\{0\}$ and is not large.

4.1 A nonprincipal coordinate

Suppose $h = h_b$ has nonprincipal image $q = h_* p$. Enumerate $\mathbb{Q} = \{r_0, r_1, \ldots\}$ without repetitions, set $s(r_n) = 2^{-n-1}$, and define $f(x) = s(h(x))$. The function is strictly positive and bounded by $1/2$. For every $\eta > 0$, its upper level set is the inverse image of a finite subset of $\mathbb{Q}$. A nonprincipal ultrafilter excludes finite sets, so $\{f < \eta\} \in p$.

Fix $T \in p$ and $x \in T^*$, and put $a = f(x) > 0$. The set $F = \{r : s(r) \geq a/2\}$ is finite. Since q is nonprincipal,

$$E_x = \{y : h(y) \notin F - h(x)\} \in p.$$

For each $\delta > 0$, choose $y \in E_x \cap (T - x) \cap (0, \delta)$. Additivity gives $h(x+y) \notin F$, hence $f(x+y) < a/2$. Lemma 3.1 applies.

Freeness of the *image* ultrafilter matters. The inverse image of a finite subset of $\mathbb{Q}$ may be infinite in $\mathbb{R}$; freeness of p alone would not justify this step.

4.2 All coordinates vanish modulo the ultrafilter

Suppose now $\ker h_b \in p$ for every b. For a fixed real x, put

$$K_x = \bigcap_{b \in \mathrm{supp}(x)} \ker h_b.$$

This is a finite intersection and belongs to p. For $y \in K_x$, the supports of x and y are disjoint.

Lemma 4.2. *For every $k \in \mathbb{N}$, the set $S_k = \{x : \ell(x) = k\}$ is not in p. Consequently $\{x : \ell(x) > m\} \in p$ for every $m \in \mathbb{N}$.*

Proof. $S_0 = \{0\}$ is not large. Suppose $k > 0$ and $S_k \in p$. For every $x \in S_k$, the large set $K_x \cap S_k$ is contained in $S_{2k} - x$. Thus $S_k \subseteq \{x : S_{2k} - x \in p\}$, so the set on the right is large. Idempotence gives $S_{2k} \in p + p = p$, contradicting disjointness of S_k and S_{2k}. This proves the first assertion. A finite union of sets excluded by an ultrafilter is excluded, yielding the second. □

Define $f(x) = 2^{-\ell(x)}$. Lemma 4.2 gives $\{f < \eta\} \in p$ for every $\eta > 0$. Given $T \in p$ and $x \in T^*$, write $a = f(x)$. For every $\delta > 0$, the set

$$(T - x) \cap (0, \delta) \cap K_x \cap \{y : \ell(y) \geq 2\}$$

is large. A member y satisfies

$$f(x + y) = 2^{-\ell(x)-\ell(y)} \leq a/4 < a/2.$$

Again Lemma 3.1 applies. These two cases prove Theorem 1.1.

5 Consequences and the scope of the answer

Corollary 5.1. *For every admissible p, $\mathcal{D}_p$ is a proper subfield containing $\mathbb{R}$ and is not order convex in $\mathbb{R}^{\mathbb{R}}/p$.*

Proof. The subfield assertion is Proposition 2.1. The main theorem gives $0 < [f]_p < 1$ with $[f]_p \notin \mathcal{D}_p$, whereas the endpoints $0, 1$ are in the domain. This contradicts order convexity. □

Although $[f]_p$ is infinitesimal, it is not zero. Equality in the ultrapower means equality on a large set, and f is nowhere zero. Consequently replacing f by the zero function is unavailable. This distinction between infinitesimal proximity and equality is essential to the example.

For comparison, the identity function represents a positive infinitesimal $\Omega = [x \mapsto x]_p$ which does belong to $\mathcal{D}_p$, with $\partial\Omega = 1$. Thus infinitesimal size alone neither includes nor excludes an element from the differentiation domain.

The answer to the source's proper-versus-full question is therefore independent of which admissible p is chosen: it is always proper. The actual subfields $\mathcal{D}_p$ may still vary. Our quantifier order is

$$\forall p \; \exists f \; \forall g \sim_p f,$$

with the additional hypotheses on p stated in the main theorem. We do not obtain a single function working for all p, nor an identification of domains for different ultrafilters.

The construction uses a Hamel basis and unrestricted functions. It does not produce Borel or measurable witnesses. Establishing properness with regular representatives, or determining which regularity classes always meet $\mathcal{D}_p$, would be a different question. The proof is also specific about its hypotheses: additive idempotence produces the sets T^* and controls rational coordinate images; concentration near zero produces arbitrarily small increments. Neither is replaced here by freeness alone.

Preparation. This is an AI-assisted hand-proof manuscript prepared for mathematical review. All arguments used for the properness theorem are included. No proof-assistant certification or established historical priority is claimed.

References

[1] S. A. Alexander and B. Dawson, *Hyperreal differentiation with an idempotent ultrafilter*, Journal of Logic and Analysis 16 (2024), paper 6. doi:10.4115/jla.2024.16.6. Source version: arXiv:2411.14689v1, 22 November 2024; especially Definitions 5–6, Theorem 9, Corollary 10 and the question following it.